00 A CONCEPT, EXPLAINED

Essential complexity.

On the model that outlives the tools – from business architecture to code.

Robert Blust · Software Engineer & Architect

01 MY THROUGH-LINE

From business architecture to code – end-to-end traceability.

Four areas shaped my career: software development, methodology, IT architecture and business architecture. I know all four.

> The goal is always end-to-end continuity – from the business domain to implementation. Only the path there was long expensive and hard.

A model captures a problem in its essential complexity – no more.

Capture the essential, leave out the unimportant. That is the real craft.

> My standard for every model, for about 15 years.

03 ONCE HARD – NOW EASIER

BEFORE

UML · SysML · your own DSL

Powerful – but hard, specialized, expensive. Few can read it.

TODAY

AI understands natural language.

With the right meta-model: the facts as Markdown – cheap to create and maintain, readable for everyone.

Same clarity – far less friction.

04 MY APPROACH

The Mental Model.

A structured knowledge base – I call it the Mental Model.

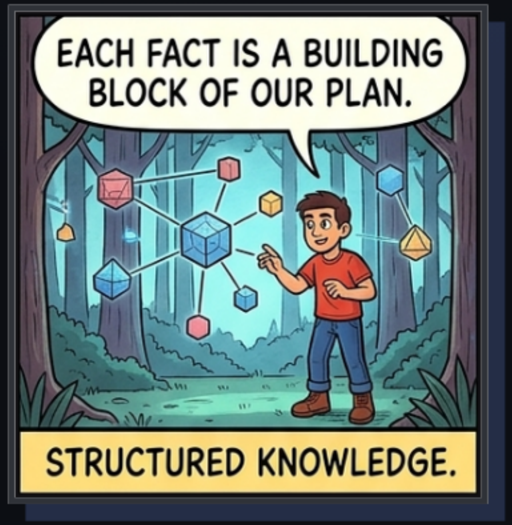
Domains & terms
Booking · Finance · Door Access · Task

How we work
Roles · teams · processes with gates

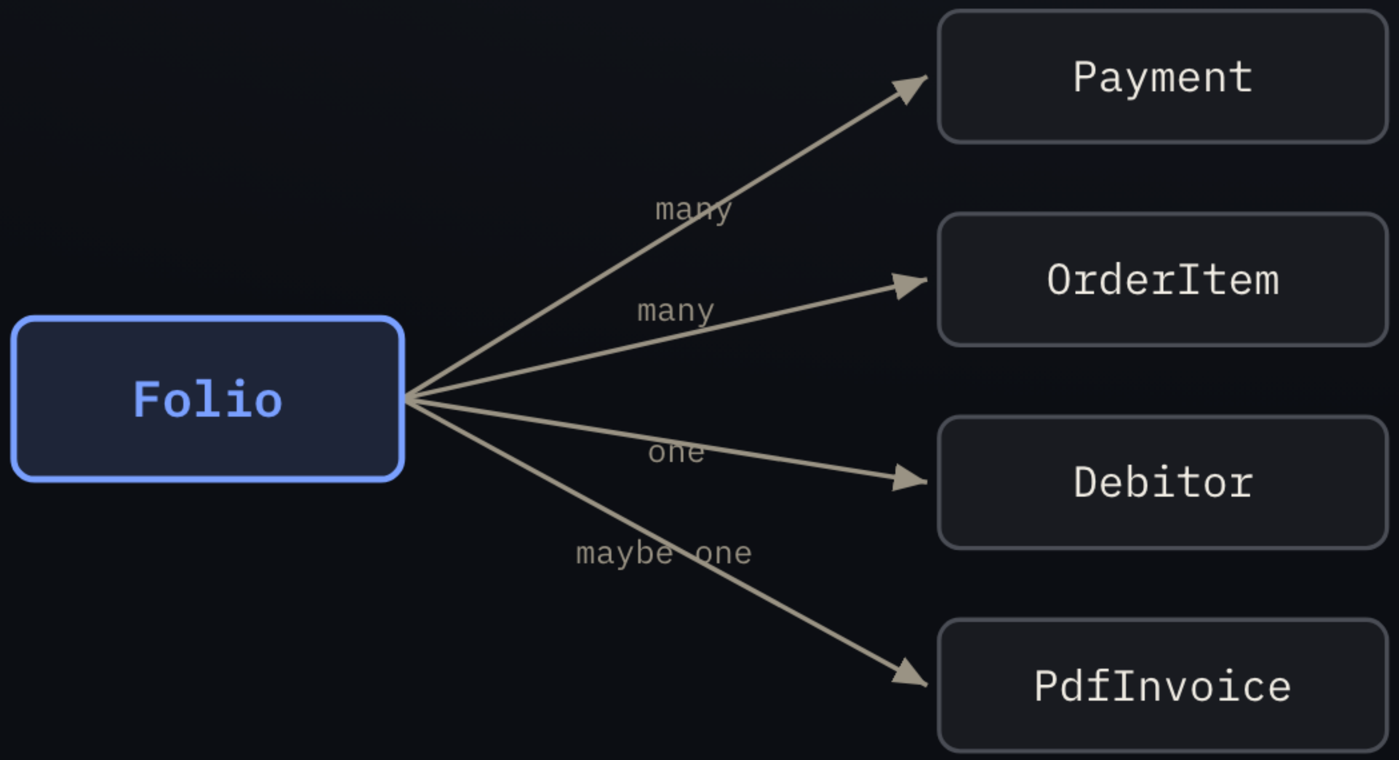
Why this way
Architecture decisions · rules

How we measure
KPIs

Facts as Markdown · structured by a meta-model · **machine-readable**



05 AN EXAMPLE – FINANCE IN A HOTEL SYSTEM



Folio
Financial account of a reservation – bundles all line items of a stay.

OrderItem
A single line item on the folio.

Payment
Payment that settles order items fully or partly.

Debitor
Who owes the bill – person or company.

PdfInvoice
The finalized invoice at check-out.

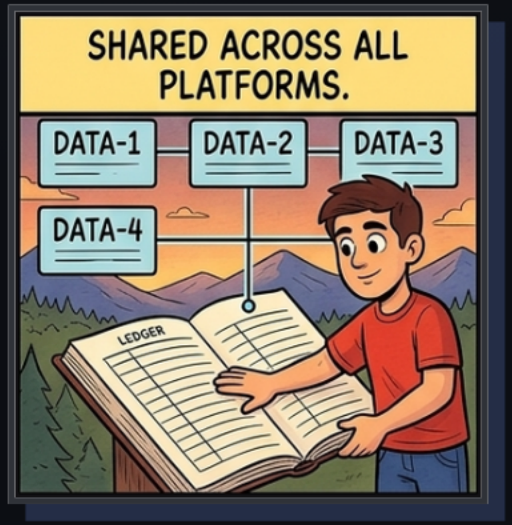
Real model from LIKE MAGIC – **Markdown + Mermaid**

Every term in one sentence – **no more, no less.**

One model that everything can be measured against.

- ✓ **Requirements** use only terms from the **glossary** – one language, no synonyms.
- ✓ **Implementation** validated against **glossary + ADRs**.
- ✓ **New terms** must land in the **glossary** first – otherwise no merge.

Not documentation but an artifact to validate against – so that **change gets cheaper and safer.**



07 WHY IT MATTERS NOW

The right context makes AI far more reliable.

> A model like the Mental Model gives AI the right context: it does not eliminate hallucinations – but it shrinks the space where AI has to guess, moving it from guessing to grounded reasoning. The cleaner the structure, the better the context. In practice: the Mental Model as a context layer for AI assistants.

The pendulum swings – back to spec-driven development.

From “no big design up front” to spec-first – because AI makes specifications fast and affordable.

> For years upfront modeling was too expensive, so agile deliberately dropped it. That math has changed.

09 TAKEAWAY

Tools come and go – **the model stays.**

The standard stays: capture essential complexity. Only now the path is open to everyone.

> That is how I still build today: close to the code, but with the model in mind. Thank you.